

Shubham Kumar Gupta

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Education

Amity University Noida, India

Bachelor of Technology in Computer Science, AI & ML Specialization

Aug 2023 – Jul 2027

- **Relevant Coursework:** Data Structures & Algorithms, Artificial Neural Networks, CNNs, RNNs, LSTMs, Computer Vision, Database Management Systems, Operating Systems, Object-Oriented Programming

Experience

WINNIIO / LifeAtlas

Sweden (Remote)

AI Engineering Intern

May 2026 – Jul 2026

- Built the initial voice-interview prototype on Retell AI / Twilio: voice in, Claude profile extraction, ranked matches via pgvector + Voyage rerank.
- Rebuilt the full voice pipeline from scratch in parallel, cutting cost ~20X vs Retell AI while keeping the same flow.
- Contributed to the ZeroClaw auth proxy; worked across edge-native AI, Digital Twin, and 3D Gaussian Splatting (3DGS) for spatial intelligence on the LifeAtlas platform.

Stakrid Logistics Private Limited

Noida, India

Founder, Lead Engineer

Jan 2025 – Jan 2026

- Built a last-mile logistics platform: 40+ REST API endpoints (Java, Spring Boot) for order tracking and delivery workflows; cut manual processing 80% and API latency from 800ms to under 200ms via query optimization and connection pooling.
- Deployed on GCP with CI/CD, load balancing, and monitoring; integrated Supabase, payment gateways, and SMS APIs with retry logic.

IET Student Chapter, Amity University

Noida, India

Technical Team Member

Mar 2024 – Jun 2025

- Built and ran technical workshops on software development and data structures for 100+ students.

Projects

Agent Reliability Stack

Jun 2026 – Present

Python, Claude API, FastAPI, Neo4j, pgvector, Voyage AI, Redis, sentence-transformers, pytest, GitHub Actions

- **Eval harness:** from-scratch CI regression gate for agent behavior — pluggable scorers (exact / regex / embedding / LLM-as-judge) plus LCS-based trajectory scoring on tool-call sequences; thread-pooled runs with failure isolation and multi-run mean±std stats.
- Calibrated the LLM-judge scorer against human-graded cases before trusting it in CI — on a 60-case benchmark: exact-match 0.867, embedding similarity 0.984, calibrated judge 1.000; wired into GitHub Actions with baseline-diffed exit codes.
- **LLM gateway:** FastAPI gateway backed by a Neo4j memory graph — one embedding call (Voyage-3.5) drives cache lookup, graph seeding, and write-back together; semantic cache (≥ 0.95 sim) returns free/instant, misses trigger a 1–2 hop graph walk (≥ 0.75 sim) that injects ranked context into the prompt.
- Auto-fallback to a secondary model tier on 429/timeout/5xx, per-call cost rows attributed to user/feature/day, and full grounding traceability (responses cite the exact context call IDs used) — Postgres + Neo4j writes happen off the request path.
- **Guardrails middleware** (in progress): pre-action validation, retry-with-repair, and a kill switch, layered in front of both systems above.

Relogged: Sandboxed Tool-Call Replay Debugger

Jul 2026

- Local-first replay debugger for LangGraph agents: monkeypatches `CompiledStateGraph` and each `ToolNode.wrap_tool_call` to record every tool call and state transition to Postgres, zero-code-change via a scoped `@record()` contextvar.
- Pure replay reconstructs a run from the DB alone — zero graph, LLM, or tool invocation; fork/override replay resumes live from LangGraph's own checkpoint history at any step with a corrected value. A sandbox test poisons the real tools to prove pure replay never touches them.

Chakra47: Autonomous OS for Physical AI

Oct 2025 – Present

- Four-layer LangGraph multi-agent swarm with neuro-symbolic governance (YOLOv8 + Open Policy Agent + SHA-256 audit trail) gating agent actions pre-execution; open-sourced, live POC on GitHub.

Military Deployment Decision System

Feb 2026

- Tactical AI analysis system — CNN-based threat detection, Claude Sonnet reasoning, hard-coded Rules of Engagement validation layer; presented at AI Impact Summit 2026 (Government of India).

Technical Skills

Languages: Python, Java, SQL, Bash

DSA: Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Recursion, Sorting, Sliding Window, Two Pointers

AI/ML: Deep Learning, Computer Vision, CNNs, RNNs, LSTMs, Transformers, RLHF, LoRA/PEFT, Hugging Face, Neuro-Symbolic AI, TensorFlow, PyTorch, YOLOv8, OpenCV, MediaPipe

Agentic & Backend: LangGraph, LangChain, Claude API, RAG, pgvector, Voyage rerank, Open Policy Agent, Spring Boot, REST API Design, JDBC, Spring Data JPA, PostgreSQL, SQLAlchemy

Cloud & DevOps: Google Cloud Platform, Docker, CI/CD, Supabase, Firebase

Concepts: System Architecture, OOP, Design Patterns, Database Design, Distributed Systems

Certifications

NCA-AIIO: AI Infrastructure & Operations Associate (NVIDIA, In Progress) · Machine Learning Specialization (Stanford/Coursera) · Java Spring Framework 6 with Spring Boot 3 (Udemy) · Build and Secure Networks in Google Cloud (Google Cloud Skills Boost)